Read the axes before you read the line

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Signed area / ES: área con signo / PT: área com sinal

Height × width = displacement. Area below zero is negative displacement.

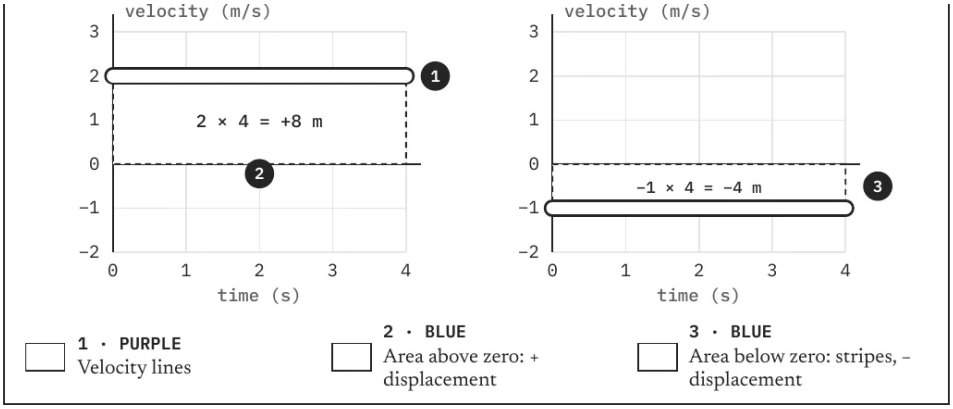
Also on your note page: Velocity–time graph, Constant velocity, Components.

B READ THE RELATIONSHIP

Signed area on a velocity–time graph gives displacement.

Displacement = signed area under a velocity–time graph

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Velocity is +3 m/s for 2 s. Find the displacement and name its direction.

Given v = +3 m/s t = 2 s

Rule $\Delta x = v \times t$

Work $\Delta x = (3 \text{ m/s})(2 \text{ s})$

Answer +6 m 6 m to the right

E SAVE FOR THE FINAL MEETING

A velocity–time graph shows whether XJ's speed is holding steady. Splitting its velocity arrow into parts shows how much is aimed at Earth and how much carries it sideways.

Read the physics: Signed area

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Read the vertical axis first. A flat line on a velocity–time graph means steady velocity. A flat line above zero means the object is moving.

02 HOW IT WORKS

Shade the region between velocity and the time axis. For steady velocity, multiply velocity by time. Add areas above zero; subtract the sizes of areas below zero. For distance, add all the sizes.

03 WHAT TO CHECK

A diagonal velocity has two components. For an angle measured from +x, use cosine for the x part and sine for the y part.

04 TRY THE EXAMPLE FROM YOUR NOTES

Velocity is +3 m/s for 2 s. Find the displacement and name its direction. +6 m 6 m to the right.

05 CONNECT TO XJ

A velocity–time graph shows whether XJ's speed is holding steady. Splitting its velocity arrow into parts shows how much is aimed at Earth and how much carries it sideways.

SAY IT BEFORE YOU WRITE

Which shaded part subtracts from displacement? The area is ___ because it is ___ zero.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Velocity–time graph

ES: gráfica velocidad-tiempo / PT: gráfico velocidade-tempo

Meaning in my words	How can you tell?
How would you explain velocity–time graph to a friend? In my words, this means ___.	Which axis helps you tell whether the cart is moving left or right? I can tell because ___.
Example or sketch	**Non-example and why**
How would you show three seconds of steady motion at −2 m/s? My example shows ___.	A student says a flat line at +2 means the cart is parked. What does that height mean? I would change ___ because ___.

Constant velocity

ES: velocidad constante / PT: velocidade constante

Meaning in my words	How can you tell?
How would you explain constant velocity to a friend? In my words, this means ___.	Photographs are one second apart. What pattern would you look for? I can tell because ___.
Example or sketch	**Non-example and why**
Draw three positions for a cart moving 2 m right each second. My example shows ___.	A bike circles a track at a steady speed. Is its velocity constant? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Signed area

ES: área con signo / PT: área com sinal

Meaning in my words	How can you tell?
How would you explain signed area to a friend? In my words, this means ___.	What do you do with the area below the time axis when you find displacement? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch a rectangle at −2 m/s lasting 3 s. What displacement does it show? My example shows ___.	Someone labels the area −2 m/s. Which graph reading did they use? I would change ___ because ___.

Components

ES: componentes / PT: componentes

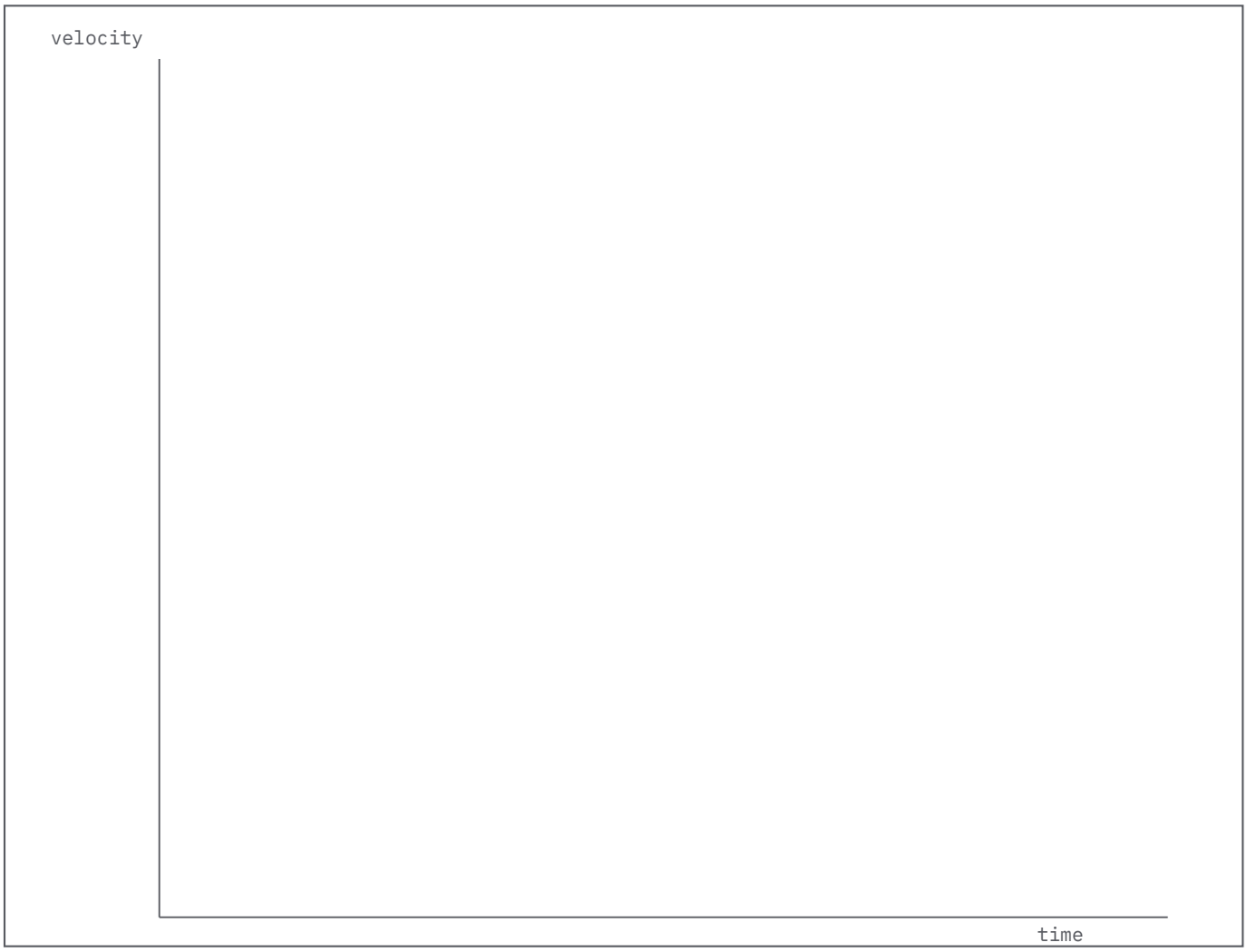
Meaning in my words	How can you tell?
What are the components of an arrow? In my words, this means ___.	If east and north are positive, which component is negative for a southeast trip? I can tell because ___.
Example or sketch	**Non-example and why**
What two arrows could you use to build a displacement of 3 m east and 4 m north? My example shows ___.	A classmate counts the diagonal and both parts as three separate trips. What is counted twice? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw and shade opposite-side rectangles. Put m/s on the vertical and s on the horizontal axis.

Labels you can use: above · below · area.



Tell a partner

The area is ___ because it is ___ zero.

On assignment: Signed area

SECTION E / BACK / FINAL MEETING ONLY Name _____

A moving camera follows a model at +3 km/s for 2 s, then −1 km/s for 2 s. The editor needs the end location and the total footage distance.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Tracking engineer**

Shade both parts of a velocity graph. Give the camera operator the final change from start, including direction.

☐ **Replay producer**

Storyboard the outbound and return shots. Tell the editor total distance and net movement; mark which part plays in reverse direction.

Word bank: above · below · area.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”